

# Collecting of SPS solvents

Date: 2023-07-17

Tags: solvent

Category: Protocol

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## Goal

Collection of dry and degassed solvents from the SPS (solvent purification system)

## Prerequisites and preparation

SPS (in ZAF 1 floor, to SPS in the labs on the left hand side)

Young-Type solvent flask

2-Neck Schlenk flask, at least one NS-14 Neck (small)

Teflon cannula

Single use needle

Molar sieve (3A)

Instruction by Alexander Meier

### Available solvents:

| SPS ZAF 107  | SPS ZAF 108        |
|--------------|--------------------|
| Cyclohexane  | Methanol           |
| Acetonitril  | Ethanol            |
| Diethylether | THF w/o stabilizer |
| DCM          | DMF                |
| THF          | Chlorbenzol        |

## Steps

| Step number | Step description                                                                                                                                                             |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <b>WORK ON THE LINE</b>                                                                                                                                                      |
| 1           | Equipp the Young type flask with a stopper and a septum. Dry the flask in the heat under vacuum three times.                                                                 |
| 2           | Add dried Molar sieve (approx. 1/5 of the total volume) to the flask. Set the flask under vacuum three times. Use a Filter frit to prevent dust going into the Schlenk line. |

|    |                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3  | Prepare the two-neck Schlenk flask by heating in vacuum for three times and after cooling to rt set the flask under argon.                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4  | Connect the two flasks via the teflon cannula using both septa. (Keep the Young type flask closed) <a href="#">set_up_preparation.jpeg</a>                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5  | In Argon counterflow: Remove the Argon Line connection of the Young type flask and close it with a GL 14 cap.                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 6  | Bring the connected flasks to the SPS, a metal clamp, gloves and paper-towels are recommended                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                            |
|    | <b>WORK ON THE SPS</b>                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7  | Start vacuum pump                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8  | Connect the free Neck of the two-neck Schlenk flask to the desired solvent of the SPS in N2 counterflow, secure the flask with a clamp <a href="#">set_up_SPS.jpeg</a>                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 9  | Open the Young type flask to the SPS. Set the flasks under vacuum and argon three times, wait till pressure shown on the SPS is constant for each cycle and then wait a few seconds.                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10 | Place the teflon cannula in the young type flask and to the bottom of the two-neck Schlenk flask.                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11 | Leave flasks under vacuum (not recommended for low boiling solvents like Et2O or DCM)                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
|    | <b>OTHER SOLVENTS</b>                                                                                                                                                                                                                                                                                                   | <b>LOW BOILING SOLVENTS (Et2O, DCM)</b>                                                                                                                                                                                                                                                                                                                                                                    |
| 12 | Open SPS to solvent till desired volume is reached. Make sure not too much solvent is transferred (You only want to fill the Young type flask, and not additionally the two neck Schlenk flask). Make sure the flow is not too high, so that solvent fills up in the two-neck flask.<br><a href="#">collecting.jpeg</a> | Set the flask under N2, then open the SPS to the desired solvent. When solvent is flowing: Put a single use needle through the septum at the Young-type solvent flask. Make sure not too much solvent is transferred (You only want to fill the Young type flask, and not additionally the two neck Schlenk flask). Make sure the flow is not too high, so that solvent fills up in the two-neck flask.13a |
| 13 |                                                                                                                                                                                                                                                                                                                         | Set the SPS to N2 and afterwards remove the needle. Remove the cannula from the Young-type flask, so that it can be closed, but is still connected to the SPS. Close the Young-type flask. Remove the flasks in N2 counterflow. Continue with step 15                                                                                                                                                      |
| 14 | Open SPS to N2 and fill the flask. Remove the cannula from the Young-type flask, so that it can be closed, but is still connected to the SPS. Close the Young-type flask. Remove the flasks in N2 counterflow                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 15 | Close the SPS. Shut down the vacuum pump. Write amount and date in the book besides the SPS                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 16 | Degas the solvent according to <a href="#">Protocol - Degassing of organic solvents</a>                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                            |

## Linked resource

Protocol - [Degassing of organic solvents in Young-type flask](#)

## Attached files

set\_up\_preparation.jpeg

sha256: dfe6c96e78dcf6c318657ffb6131aa9113fbbc2469c83298c97338a954f14346



collecting.jpeg

sha256: 0cc3145aa998d4f54e7b1ea07bd11dc96ce5860cbb5b0206a2972bf3ef2a9bba



set\_up\_SPS.jpeg

sha256: 3ad6a1e8bcf318a2a8bc93ca373337f0f765165fa0871bf64c40135745f2cef3



Unique eLabID: 20230717-cf786db2b82eb53113d1fdfedf363f653c0a81cf  
Link: <https://elab.water-splitting.org/database.php?mode=view&id=66>